

5

INTEGRATION



Figure 5.1 Iceboating is a popular winter sport in parts of the northern United States and Europe. (credit: modification of work by Carter Brown, Flickr)

Chapter Outline

- 5.1 Approximating Areas
- 5.2 The Definite Integral
- 5.3 The Fundamental Theorem of Calculus
- 5.4 Integration Formulas and the Net Change Theorem
- 5.5 Substitution
- 5.6 Integrals Involving Exponential and Logarithmic Functions
- 5.7 Integrals Resulting in Inverse Trigonometric Functions



Introduction

Iceboats are a common sight on the lakes of Wisconsin and Minnesota on winter weekends. Iceboats are similar to sailboats, but they are fitted with runners, or “skates,” and are designed to run over the ice, rather than on water. Iceboats can move very quickly, and many ice boating enthusiasts are drawn to the sport because of the speed. Top iceboat racers can attain speeds up to five times the wind speed. If we know how fast an iceboat is moving, we can use integration to determine how far it travels. We revisit this question later in the chapter (see [Example 5.27](#)).

Determining distance from velocity is just one of many applications of integration. In fact, integrals are used in a wide variety of mechanical and physical applications. In this chapter, we first introduce the theory behind integration and use

integrals to calculate areas. From there, we develop the Fundamental Theorem of Calculus, which relates differentiation and integration. We then study some basic integration techniques and briefly examine some applications.

5.1 Approximating Areas

Learning Objectives

5.1.1 Use sigma (summation) notation to calculate sums and powers of integers.

5.1.2 Use the sum of rectangular areas to approximate the area under a curve.

5.1.3 Use Riemann sums to approximate area.

Archimedes was fascinated with calculating the areas of various shapes—in other words, the amount of space enclosed by the shape. He used a process that has come to be known as the *method of exhaustion*, which used smaller and smaller shapes, the areas of which could be calculated exactly, to fill an irregular region and thereby obtain closer and closer approximations to the total area. In this process, an area bounded by curves is filled with rectangles, triangles, and shapes with exact area formulas. These areas are then summed to approximate the area of the curved region.

In this section, we develop techniques to approximate the area between a curve, defined by a function $f(x)$, and the x -axis on a closed interval $[a, b]$. Like Archimedes, we first approximate the area under the curve using shapes of known area (namely, rectangles). By using smaller and smaller rectangles, we get closer and closer approximations to the area. Taking a limit allows us to calculate the exact area under the curve.

Let's start by introducing some notation to make the calculations easier. We then consider the case when $f(x)$ is continuous and nonnegative. Later in the chapter, we relax some of these restrictions and develop techniques that apply in more general cases.

Sigma (Summation) Notation

As mentioned, we will use shapes of known area to approximate the area of an irregular region bounded by curves. This process often requires adding up long strings of numbers. To make it easier to write down these lengthy sums, we look at some new notation here, called **sigma notation** (also known as **summation notation**). The Greek capital letter Σ , sigma, is used to express long sums of values in a compact form. For example, if we want to add all the integers from 1 to 20 without sigma notation, we have to write

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 11 + 12 + 13 + 14 + 15 + 16 + 17 + 18 + 19 + 20.$$

We could probably skip writing a couple of terms and write

$$1 + 2 + 3 + 4 + \cdots + 19 + 20,$$

which is better, but still cumbersome. With sigma notation, we write this sum as

$$\sum_{i=1}^{20} i,$$

which is much more compact.

Typically, sigma notation is presented in the form

$$\sum_{i=1}^n a_i$$

where a_i describes the terms to be added, and the i is called the *index*. Each term is evaluated, then we sum all the values, beginning with the value when $i = 1$ and ending with the value when $i = n$. For example, an expression like

$\sum_{i=2}^7 s_i$ is interpreted as $s_2 + s_3 + s_4 + s_5 + s_6 + s_7$. Note that the index is used only to keep track of the terms to be

added; it does not factor into the calculation of the sum itself. The index is therefore called a *dummy variable*. We can use any letter we like for the index. Typically, mathematicians use i , j , k , m , and n for indices.

Let's try a couple of examples of using sigma notation.

EXAMPLE 5.1

Using Sigma Notation

- a. Write in sigma notation and evaluate the sum of terms 3^i for $i = 1, 2, 3, 4, 5$.

b. Write the sum in sigma notation:

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25}.$$

 Solution

a. Write

$$\begin{aligned}\sum_{i=1}^5 3^i &= 3 + 3^2 + 3^3 + 3^4 + 3^5 \\ &= 363.\end{aligned}$$

b. The denominator of each term is a perfect square. Using sigma notation, this sum can be written as $\sum_{i=1}^5 \frac{1}{i^2}$.

 5.1 Write in sigma notation and evaluate the sum of terms 2^i for $i = 3, 4, 5, 6$.

The properties associated with the summation process are given in the following rule.

Rule: Properties of Sigma Notation

Let $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ represent two sequences of terms and let c be a constant. The following properties hold for all positive integers n and for integers m , with $1 \leq m \leq n$.

1.

$$\sum_{i=1}^n c = nc \quad (5.1)$$

2.

$$\sum_{i=1}^n ca_i = c \sum_{i=1}^n a_i \quad (5.2)$$

3.

$$\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i \quad (5.3)$$

4.

$$\sum_{i=1}^n (a_i - b_i) = \sum_{i=1}^n a_i - \sum_{i=1}^n b_i \quad (5.4)$$

5.

$$\sum_{i=1}^n a_i = \sum_{i=1}^m a_i + \sum_{i=m+1}^n a_i \quad (5.5)$$

Proof

We prove properties 2. and 3. here, and leave proof of the other properties to the Exercises.

2. We have

$$\begin{aligned}\sum_{i=1}^n ca_i &= ca_1 + ca_2 + ca_3 + \cdots + ca_n \\ &= c(a_1 + a_2 + a_3 + \cdots + a_n)\end{aligned}$$

$$= c \sum_{i=1}^n a_i.$$

3. We have

$$\begin{aligned}\sum_{i=1}^n (a_i + b_i) &= (a_1 + b_1) + (a_2 + b_2) + (a_3 + b_3) + \cdots + (a_n + b_n) \\ &= (a_1 + a_2 + a_3 + \cdots + a_n) + (b_1 + b_2 + b_3 + \cdots + b_n)\end{aligned}$$

$$= \sum_{i=1}^n a_i + \sum_{i=1}^n b_i.$$

□

A few more formulas for frequently found functions simplify the summation process further. These are shown in the next rule, for **sums and powers of integers**, and we use them in the next set of examples.

Rule: Sums and Powers of Integers

1. The sum of n integers is given by

$$\sum_{i=1}^n i = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

2. The sum of consecutive integers squared is given by

$$\sum_{i=1}^n i^2 = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

3. The sum of consecutive integers cubed is given by

$$\sum_{i=1}^n i^3 = 1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}.$$

EXAMPLE 5.2

Evaluation Using Sigma Notation

Write using sigma notation and evaluate:

- The sum of the terms $(i - 3)^2$ for $i = 1, 2, \dots, 200$.
- The sum of the terms $(i^3 - i^2)$ for $i = 1, 2, 3, 4, 5, 6$.

Solution

- Multiplying out $(i - 3)^2$, we can break the expression into three terms.

$$\begin{aligned}
\sum_{i=1}^{200} (i-3)^2 &= \sum_{i=1}^{200} (i^2 - 6i + 9) \\
&= \sum_{i=1}^{200} i^2 - \sum_{i=1}^{200} 6i + \sum_{i=1}^{200} 9 \\
&= \sum_{i=1}^{200} i^2 - 6 \sum_{i=1}^{200} i + \sum_{i=1}^{200} 9 \\
&= \frac{200(200+1)(400+1)}{6} - 6 \left[\frac{200(200+1)}{2} \right] + 9(200) \\
&= 2,686,700 - 120,600 + 1800 \\
&= 2,567,900
\end{aligned}$$

b. Use sigma notation property iv. and the rules for the sum of squared terms and the sum of cubed terms.

$$\begin{aligned}
\sum_{i=1}^6 (i^3 - i^2) &= \sum_{i=1}^6 i^3 - \sum_{i=1}^6 i^2 \\
&= \frac{6^2(6+1)^2}{4} - \frac{6(6+1)(2(6)+1)}{6} \\
&= \frac{1764}{4} - \frac{546}{6} \\
&= 350
\end{aligned}$$

✓ 5.2 Find the sum of the values of $4 + 3i$ for $i = 1, 2, \dots, 100$.

EXAMPLE 5.3

Finding the Sum of the Function Values

Find the sum of the values of $f(x) = x^3$ over the integers $1, 2, 3, \dots, 10$.

✓ Solution

Using the formula, we have

$$\begin{aligned}
\sum_{i=1}^{10} i^3 &= \frac{(10)^2(10+1)^2}{4} \\
&= \frac{100(121)}{4} \\
&= 3025.
\end{aligned}$$

✓ 5.3 Evaluate the sum indicated by the notation $\sum_{k=1}^{20} (2k + 1)$.

Approximating Area

Now that we have the necessary notation, we return to the problem at hand: approximating the area under a curve. Let $f(x)$ be a continuous, nonnegative function defined on the closed interval $[a, b]$. We want to approximate the area A bounded by $f(x)$ above, the x -axis below, the line $x = a$ on the left, and the line $x = b$ on the right (Figure 5.2).

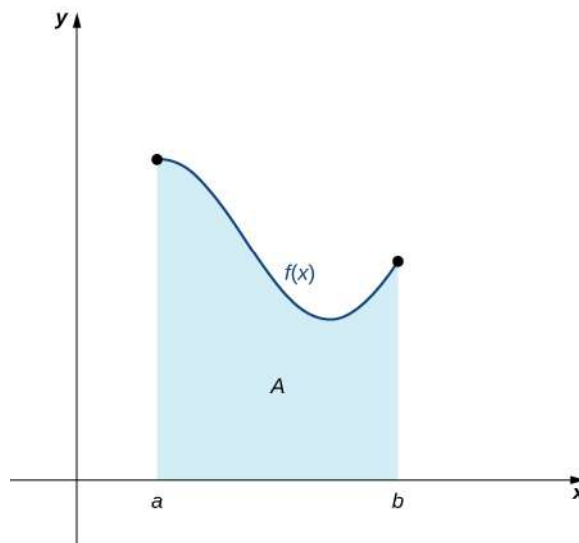


Figure 5.2 An area (shaded region) bounded by the curve $f(x)$ at top, the x -axis at bottom, the line $x = a$ to the left, and the line $x = b$ at right.

How do we approximate the area under this curve? The approach is a geometric one. By dividing a region into many small shapes that have known area formulas, we can sum these areas and obtain a reasonable estimate of the true area. We begin by dividing the interval $[a, b]$ into n subintervals of equal width, $\frac{b-a}{n}$. We do this by selecting equally spaced points $x_0, x_1, x_2, \dots, x_n$ with $x_0 = a, x_n = b$, and

$$x_i - x_{i-1} = \frac{b-a}{n}$$

for $i = 1, 2, 3, \dots, n$.

We denote the width of each subinterval with the notation Δx , so $\Delta x = \frac{b-a}{n}$ and

$$x_i = x_0 + i\Delta x$$

for $i = 1, 2, 3, \dots, n$. This notion of dividing an interval $[a, b]$ into subintervals by selecting points from within the interval is used quite often in approximating the area under a curve, so let's define some relevant terminology.

Definition

A set of points $P = \{x_i\}$ for $i = 0, 1, 2, \dots, n$ with $a = x_0 < x_1 < x_2 < \dots < x_n = b$, which divides the interval $[a, b]$ into subintervals of the form $[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n]$ is called a **partition** of $[a, b]$. If the subintervals all have the same width, the set of points forms a **regular partition** of the interval $[a, b]$.

We can use this regular partition as the basis of a method for estimating the area under the curve. We next examine two methods: the left-endpoint approximation and the right-endpoint approximation.

Rule: Left-Endpoint Approximation

On each subinterval $[x_{i-1}, x_i]$ (for $i = 1, 2, 3, \dots, n$), construct a rectangle with width Δx and height equal to $f(x_{i-1})$, which is the function value at the left endpoint of the subinterval. Then the area of this rectangle is $f(x_{i-1}) \Delta x$. Adding the areas of all these rectangles, we get an approximate value for A (Figure 5.3). We use the notation L_n to denote that this is a **left-endpoint approximation** of A using n subintervals.

$$A \approx L_n = f(x_0) \Delta x + f(x_1) \Delta x + \cdots + f(x_{n-1}) \Delta x$$

(5.6)

$$= \sum_{i=1}^n f(x_{i-1}) \Delta x$$

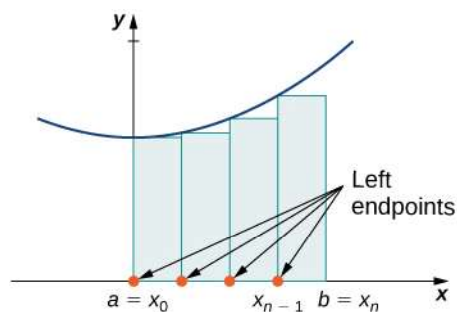


Figure 5.3 In the left-endpoint approximation of area under a curve, the height of each rectangle is determined by the function value at the left of each subinterval.

The second method for approximating area under a curve is the right-endpoint approximation. It is almost the same as the left-endpoint approximation, but now the heights of the rectangles are determined by the function values at the right of each subinterval.

Rule: Right-Endpoint Approximation

Construct a rectangle on each subinterval $[x_{i-1}, x_i]$, only this time the height of the rectangle is determined by the function value $f(x_i)$ at the right endpoint of the subinterval. Then, the area of each rectangle is $f(x_i) \Delta x$ and the approximation for A is given by

$$A \approx R_n = f(x_1) \Delta x + f(x_2) \Delta x + \cdots + f(x_n) \Delta x$$

(5.7)

$$= \sum_{i=1}^n f(x_i) \Delta x.$$

The notation R_n indicates this is a **right-endpoint approximation** for A (Figure 5.4).

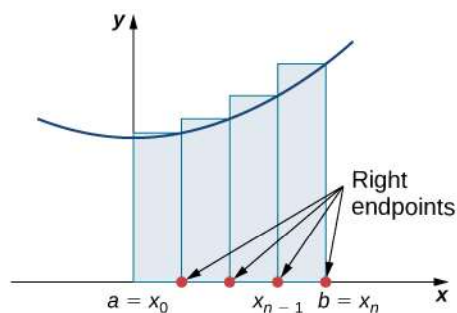


Figure 5.4 In the right-endpoint approximation of area under a curve, the height of each rectangle is determined by the function value at the right of each subinterval. Note that the right-endpoint approximation differs from the left-endpoint approximation in Figure 5.3.

The graphs in Figure 5.5 represent the curve $f(x) = \frac{x^2}{2}$. In graph (a) we divide the region represented by the interval

$[0, 3]$ into six subintervals, each of width 0.5. Thus, $\Delta x = 0.5$. We then form six rectangles by drawing vertical lines perpendicular to x_{i-1} , the left endpoint of each subinterval. We determine the height of each rectangle by calculating $f(x_{i-1})$ for $i = 1, 2, 3, 4, 5, 6$. The intervals are $[0, 0.5]$, $[0.5, 1]$, $[1, 1.5]$, $[1.5, 2]$, $[2, 2.5]$, $[2.5, 3]$. We find the area of each rectangle by multiplying the height by the width. Then, the sum of the rectangular areas approximates the area between $f(x)$ and the x -axis. When the left endpoints are used to calculate height, we have a left-endpoint approximation. Thus,

$$\begin{aligned} A \approx L_6 &= \sum_{i=1}^6 f(x_{i-1}) \Delta x = f(x_0) \Delta x + f(x_1) \Delta x + f(x_2) \Delta x + f(x_3) \Delta x + f(x_4) \Delta x + f(x_5) \Delta x \\ &= f(0) 0.5 + f(0.5) 0.5 + f(1) 0.5 + f(1.5) 0.5 + f(2) 0.5 + f(2.5) 0.5 \\ &= (0) 0.5 + (0.125) 0.5 + (0.5) 0.5 + (1.125) 0.5 + (2) 0.5 + (3.125) 0.5 \\ &= 0 + 0.0625 + 0.25 + 0.5625 + 1 + 1.5625 \\ &= 3.4375. \end{aligned}$$

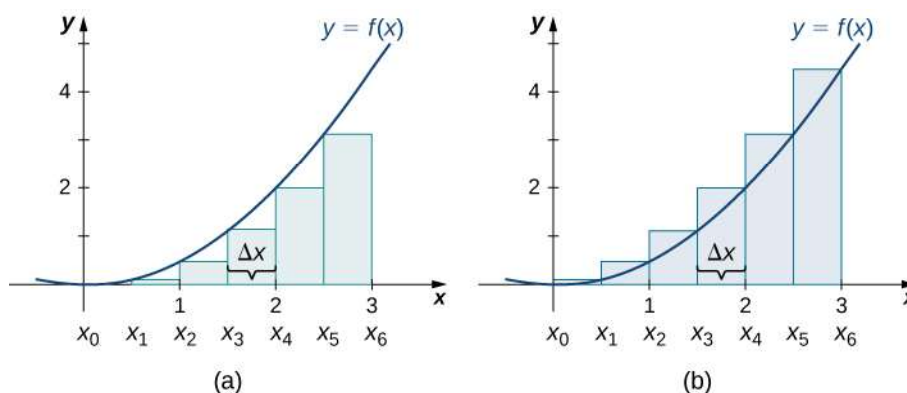


Figure 5.5 Methods of approximating the area under a curve by using (a) the left endpoints and (b) the right endpoints.

In [Figure 5.5\(b\)](#), we draw vertical lines perpendicular to x_i such that x_i is the right endpoint of each subinterval, and calculate $f(x_i)$ for $i = 1, 2, 3, 4, 5, 6$. We multiply each $f(x_i)$ by Δx to find the rectangular areas, and then add them. This is a right-endpoint approximation of the area under $f(x)$. Thus,

$$\begin{aligned} A \approx R_6 &= \sum_{i=1}^6 f(x_i) \Delta x = f(x_1) \Delta x + f(x_2) \Delta x + f(x_3) \Delta x + f(x_4) \Delta x + f(x_5) \Delta x + f(x_6) \Delta x \\ &= f(0.5) 0.5 + f(1) 0.5 + f(1.5) 0.5 + f(2) 0.5 + f(2.5) 0.5 + f(3) 0.5 \\ &= (0.125) 0.5 + (0.5) 0.5 + (1.125) 0.5 + (2) 0.5 + (3.125) 0.5 + (4.5) 0.5 \\ &= 0.0625 + 0.25 + 0.5625 + 1 + 1.5625 + 2.25 \\ &= 5.6875. \end{aligned}$$

EXAMPLE 5.4

Approximating the Area Under a Curve

Use both left-endpoint and right-endpoint approximations to approximate the area under the curve of $f(x) = x^2$ on the interval $[0, 2]$; use $n = 4$.

Solution

First, divide the interval $[0, 2]$ into n equal subintervals. Using $n = 4$, $\Delta x = \frac{(2-0)}{4} = 0.5$. This is the width of each rectangle. The intervals $[0, 0.5]$, $[0.5, 1]$, $[1, 1.5]$, $[1.5, 2]$ are shown in [Figure 5.6](#). Using a left-endpoint approximation, the heights are $f(0) = 0$, $f(0.5) = 0.25$, $f(1) = 1$, $f(1.5) = 2.25$. Then,

$$\begin{aligned}
 L_4 &= f(x_0) \Delta x + f(x_1) \Delta x + f(x_2) \Delta x + f(x_3) \Delta x \\
 &= 0(0.5) + 0.25(0.5) + 1(0.5) + 2.25(0.5) \\
 &= 1.75.
 \end{aligned}$$

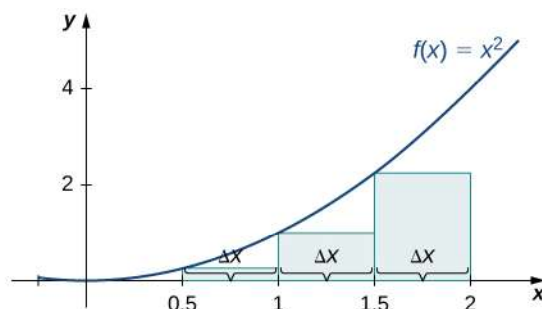


Figure 5.6 The graph shows the left-endpoint approximation of the area under $f(x) = x^2$ from 0 to 2.

The right-endpoint approximation is shown in [Figure 5.7](#). The intervals are the same, $\Delta x = 0.5$, but now use the right endpoint to calculate the height of the rectangles. We have

$$\begin{aligned}
 R_4 &= f(x_1) \Delta x + f(x_2) \Delta x + f(x_3) \Delta x + f(x_4) \Delta x \\
 &= 0.25(0.5) + 1(0.5) + 2.25(0.5) + 4(0.5) \\
 &= 3.75.
 \end{aligned}$$

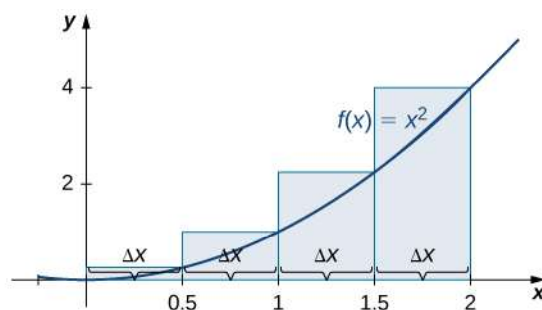


Figure 5.7 The graph shows the right-endpoint approximation of the area under $f(x) = x^2$ from 0 to 2.

The left-endpoint approximation is 1.75; the right-endpoint approximation is 3.75.

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- ✓ 5.4 Sketch left-endpoint and right-endpoint approximations for $f(x) = \frac{1}{x}$ on $[1, 2]$; use $n = 4$. Approximate the area using both methods.

Looking at [Figure 5.5](#) and the graphs in [Example 5.4](#), we can see that when we use a small number of intervals, neither the left-endpoint approximation nor the right-endpoint approximation is a particularly accurate estimate of the area under the curve. However, it seems logical that if we increase the number of points in our partition, our estimate of A will improve. We will have more rectangles, but each rectangle will be thinner, so we will be able to fit the rectangles to the curve more precisely.

We can demonstrate the improved approximation obtained through smaller intervals with an example. Let's explore the idea of increasing n , first in a left-endpoint approximation with four rectangles, then eight rectangles, and finally 32 rectangles. Then, let's do the same thing in a right-endpoint approximation, using the same sets of intervals, of the same curved region. [Figure 5.8](#) shows the area of the region under the curve $f(x) = (x - 1)^3 + 4$ on the interval $[0, 2]$ using a left-endpoint approximation where $n = 4$. The width of each rectangle is

$$\Delta x = \frac{2 - 0}{4} = \frac{1}{2}.$$

The area is approximated by the summed areas of the rectangles, or

$$\begin{aligned}
 L_4 &= f(0)(0.5) + f(0.5)(0.5) + f(1)(0.5) + f(1.5)(0.5) \\
 &= 7.5.
 \end{aligned}$$

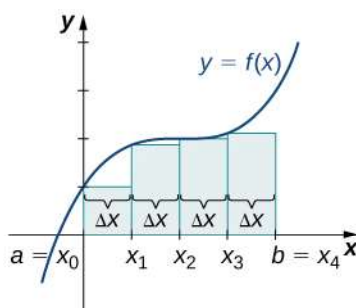


Figure 5.8 With a left-endpoint approximation and dividing the region from a to b into four equal intervals, the area under the curve is approximately equal to the sum of the areas of the rectangles.

[Figure 5.9](#) shows the same curve divided into eight subintervals. Comparing the graph with four rectangles in [Figure 5.8](#) with this graph with eight rectangles, we can see there appears to be less white space under the curve when $n = 8$. This white space is area under the curve we are unable to include using our approximation. The area of the rectangles is

$$\begin{aligned} L_8 &= f(0)(0.25) + f(0.25)(0.25) + f(0.5)(0.25) + f(0.75)(0.25) \\ &\quad + f(1)(0.25) + f(1.25)(0.25) + f(1.5)(0.25) + f(1.75)(0.25) \\ &= 7.75. \end{aligned}$$

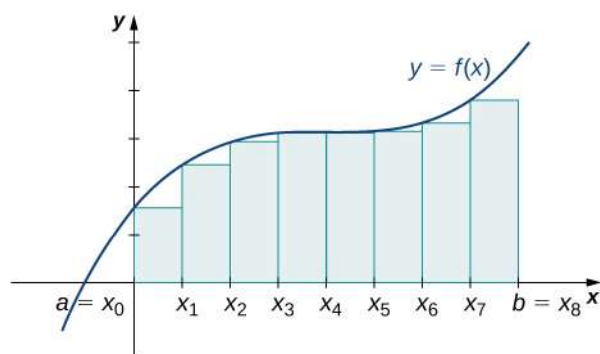


Figure 5.9 The region under the curve is divided into $n = 8$ rectangular areas of equal width for a left-endpoint approximation.

The graph in [Figure 5.10](#) shows the same function with 32 rectangles inscribed under the curve. There appears to be little white space left. The area occupied by the rectangles is

$$\begin{aligned} L_{32} &= f(0)(0.0625) + f(0.0625)(0.0625) + f(0.125)(0.0625) + \cdots + f(1.9375)(0.0625) \\ &= 7.9375. \end{aligned}$$

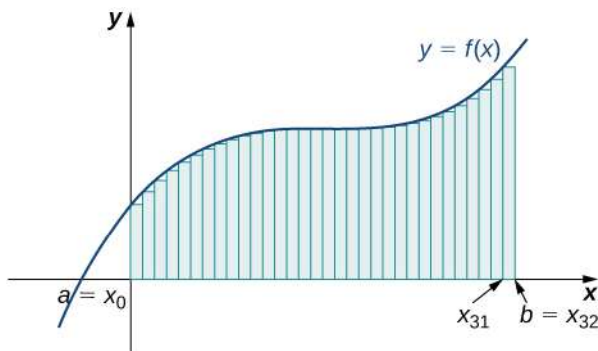


Figure 5.10 Here, 32 rectangles are inscribed under the curve for a left-endpoint approximation.

We can carry out a similar process for the right-endpoint approximation method. A right-endpoint approximation of the same curve, using four rectangles ([Figure 5.11](#)), yields an area

$$\begin{aligned} R_4 &= f(0.5)(0.5) + f(1)(0.5) + f(1.5)(0.5) + f(2)(0.5) \\ &= 8.5. \end{aligned}$$

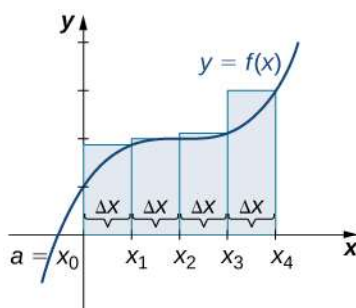


Figure 5.11 Now we divide the area under the curve into four equal subintervals for a right-endpoint approximation.

Dividing the region over the interval $[0, 2]$ into eight rectangles results in $\Delta x = \frac{2-0}{8} = 0.25$. The graph is shown in [Figure 5.12](#). The area is

$$\begin{aligned} R_8 &= f(0.25)(0.25) + f(0.5)(0.25) + f(0.75)(0.25) + f(1)(0.25) \\ &\quad + f(1.25)(0.25) + f(1.5)(0.25) + f(1.75)(0.25) + f(2)(0.25) \\ &= 8.25. \end{aligned}$$

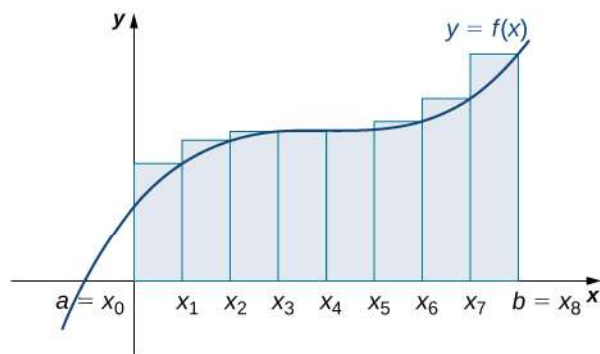


Figure 5.12 Here we use right-endpoint approximation for a region divided into eight equal subintervals.

Last, the right-endpoint approximation with $n = 32$ is close to the actual area ([Figure 5.13](#)). The area is approximately

$$\begin{aligned} R_{32} &= f(0.0625)(0.0625) + f(0.125)(0.0625) + f(0.1875)(0.0625) + \cdots + f(2)(0.0625) \\ &= 8.0625. \end{aligned}$$

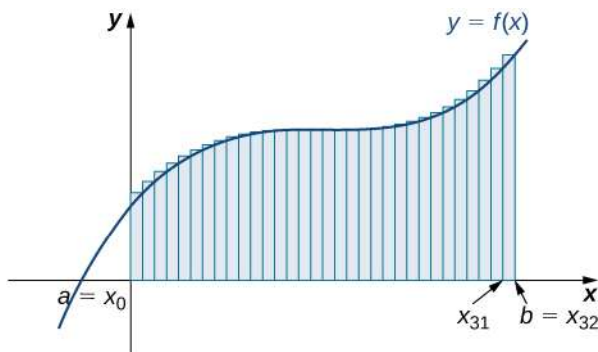


Figure 5.13 The region is divided into 32 equal subintervals for a right-endpoint approximation.

Based on these figures and calculations, it appears we are on the right track; the rectangles appear to approximate the area under the curve better as n gets larger. Furthermore, as n increases, both the left-endpoint and right-endpoint approximations appear to approach an area of 8 square units. [Table 5.1](#) shows a numerical comparison of the left- and right-endpoint methods. The idea that the approximations of the area under the curve get better and better as n gets larger and larger is very important, and we now explore this idea in more detail.

Values of n	Approximate Area L_n	Approximate Area R_n
$n = 4$	7.5	8.5
$n = 8$	7.75	8.25
$n = 32$	7.94	8.06

Table 5.1 Converging Values of Left- and Right-Endpoint Approximations as n Increases

Forming Riemann Sums

So far we have been using rectangles to approximate the area under a curve. The heights of these rectangles have been determined by evaluating the function at either the right or left endpoints of the subinterval $[x_{i-1}, x_i]$. In reality, there is no reason to restrict evaluation of the function to one of these two points only. We could evaluate the function at any point x_i^* in the subinterval $[x_{i-1}, x_i]$, and use $f(x_i^*)$ as the height of our rectangle. This gives us an estimate for the area of the form

$$A \approx \sum_{i=1}^n f(x_i^*) \Delta x.$$

A sum of this form is called a Riemann sum, named for the 19th-century mathematician Bernhard Riemann, who developed the idea.

Definition

Let $f(x)$ be defined on a closed interval $[a, b]$ and let P be a regular partition of $[a, b]$. Let Δx be the width of each subinterval $[x_{i-1}, x_i]$ and for each i , let x_i^* be any point in $[x_{i-1}, x_i]$. A **Riemann sum** is defined for $f(x)$ as

$$\sum_{i=1}^n f(x_i^*) \Delta x.$$

Recall that with the left- and right-endpoint approximations, the estimates seem to get better and better as n get larger and larger. The same thing happens with Riemann sums. Riemann sums give better approximations for larger values of n . We are now ready to define the area under a curve in terms of Riemann sums.

Definition

Let $f(x)$ be a continuous, nonnegative function on an interval $[a, b]$, and let $\sum_{i=1}^n f(x_i^*) \Delta x$ be a Riemann sum for $f(x)$. Then, the **area under the curve** $y = f(x)$ on $[a, b]$ is given by

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x.$$

MEDIA

See a [graphical demonstration \(http://www.openstax.org/l/20_riemannsums\)](http://www.openstax.org/l/20_riemannsums) of the construction of a Riemann sum.

Some subtleties here are worth discussing. First, note that taking the limit of a sum is a little different from taking the limit of a function $f(x)$ as x goes to infinity. Limits of sums are discussed in detail in the chapter on [Sequences and Series \(http://openstax.org/books/calculus-volume-2/pages/5-introduction\)](http://openstax.org/books/calculus-volume-2/pages/5-introduction); however, for now we can assume that the computational techniques we used to compute limits of functions can also be used to calculate limits of sums.

Second, we must consider what to do if the expression converges to different limits for different choices of $\{x_i^*\}$. Fortunately, this does not happen. Although the proof is beyond the scope of this text, it can be shown that if $f(x)$ is continuous on the closed interval $[a, b]$, then $\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$ exists and is unique (in other words, it does not depend on the choice of $\{x_i^*\}$).

We look at some examples shortly. But, before we do, let's take a moment and talk about some specific choices for $\{x_i^*\}$. Although any choice for $\{x_i^*\}$ gives us an estimate of the area under the curve, we don't necessarily know whether that estimate is too high (overestimate) or too low (underestimate). If it is important to know whether our estimate is high or low, we can select our value for $\{x_i^*\}$ to guarantee one result or the other.

If we want an overestimate, for example, we can choose $\{x_i^*\}$ such that for $i = 1, 2, 3, \dots, n$, $f(x_i^*) \geq f(x)$ for all $x \in [x_{i-1}, x_i]$. In other words, we choose $\{x_i^*\}$ so that for $i = 1, 2, 3, \dots, n$, $f(x_i^*)$ is the maximum function value on the interval $[x_{i-1}, x_i]$. If we select $\{x_i^*\}$ in this way, then the Riemann sum $\sum_{i=1}^n f(x_i^*) \Delta x$ is called an **upper sum**.

Similarly, if we want an underestimate, we can choose $\{x_i^*\}$ so that for $i = 1, 2, 3, \dots, n$, $f(x_i^*)$ is the minimum function value on the interval $[x_{i-1}, x_i]$. In this case, the associated Riemann sum is called a **lower sum**. Note that if $f(x)$ is either increasing or decreasing throughout the interval $[a, b]$, then the maximum and minimum values of the function occur at the endpoints of the subintervals, so the upper and lower sums are just the same as the left- and right-endpoint approximations.

EXAMPLE 5.5

Finding Lower and Upper Sums

Find a lower sum for $f(x) = 10 - x^2$ on $[1, 2]$; let $n = 4$ subintervals.

Solution

With $n = 4$ over the interval $[1, 2]$, $\Delta x = \frac{1}{4}$. We can list the intervals as $[1, 1.25]$, $[1.25, 1.5]$, $[1.5, 1.75]$, $[1.75, 2]$.

Because the function is decreasing over the interval $[1, 2]$, [Figure 5.14](#) shows that a lower sum is obtained by using the right endpoints.

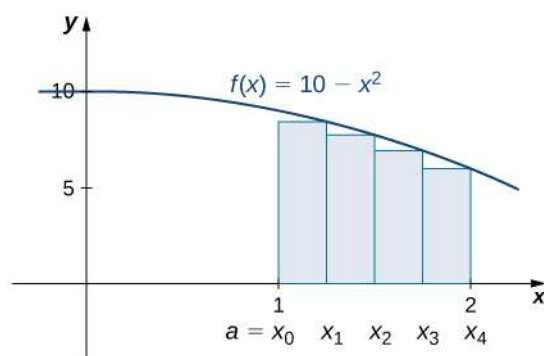


Figure 5.14 The graph of $f(x) = 10 - x^2$ is set up for a right-endpoint approximation of the area bounded by the curve and the x -axis on $[1, 2]$, and it shows a lower sum.

The Riemann sum is

$$\begin{aligned} \sum_{k=1}^4 (10 - x_k^2) (0.25) &= 0.25 [10 - (1.25)^2 + 10 - (1.5)^2 + 10 - (1.75)^2 + 10 - (2)^2] \\ &= 0.25 [8.4375 + 7.75 + 6.9375 + 6] \\ &= 7.28. \end{aligned}$$

The area of 7.28 is a lower sum and an underestimate.

-  5.5 a. Find an upper sum for $f(x) = 10 - x^2$ on $[1, 2]$; let $n = 4$.
b. Sketch the approximation.

EXAMPLE 5.6**Finding Lower and Upper Sums for $f(x) = \sin x$**

Find a lower sum for $f(x) = \sin x$ over the interval $[a, b] = [0, \frac{\pi}{2}]$; let $n = 6$.

✓ Solution

Let's first look at the graph in [Figure 5.15](#) to get a better idea of the area of interest.

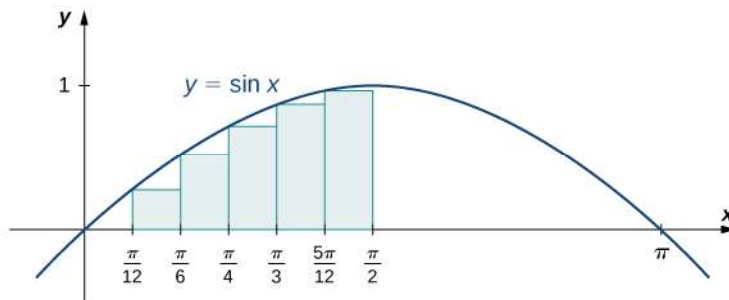


Figure 5.15 The graph of $y = \sin x$ is divided into six regions: $\Delta x = \frac{\pi/2}{6} = \frac{\pi}{12}$.

The intervals are $[0, \frac{\pi}{12}]$, $[\frac{\pi}{12}, \frac{\pi}{6}]$, $[\frac{\pi}{6}, \frac{\pi}{4}]$, $[\frac{\pi}{4}, \frac{\pi}{3}]$, $[\frac{\pi}{3}, \frac{5\pi}{12}]$, and $[\frac{5\pi}{12}, \frac{\pi}{2}]$. Note that $f(x) = \sin x$ is increasing on the interval $[0, \frac{\pi}{2}]$, so a left-endpoint approximation gives us the lower sum. A left-endpoint approximation is the Riemann

sum $\sum_{i=0}^5 \sin x_i \left(\frac{\pi}{12} \right)$. We have

$$\begin{aligned} A &\approx \sin(0) \left(\frac{\pi}{12} \right) + \sin \left(\frac{\pi}{12} \right) \left(\frac{\pi}{12} \right) + \sin \left(\frac{\pi}{6} \right) \left(\frac{\pi}{12} \right) + \sin \left(\frac{\pi}{4} \right) \left(\frac{\pi}{12} \right) + \sin \left(\frac{\pi}{3} \right) \left(\frac{\pi}{12} \right) + \sin \left(\frac{5\pi}{12} \right) \left(\frac{\pi}{12} \right) \\ &= 0.863. \end{aligned}$$

- ✓ 5.6 Using the function $f(x) = \sin x$ over the interval $[0, \frac{\pi}{2}]$, find an upper sum; let $n = 6$.

**SECTION 5.1 EXERCISES**

1. State whether the given sums are equal or unequal.

- $\sum_{i=1}^{10} i$ and $\sum_{k=1}^{10} k$
- $\sum_{i=1}^{10} i$ and $\sum_{i=6}^{15} (i - 5)$
- $\sum_{i=1}^{10} i(i - 1)$ and $\sum_{j=0}^9 (j + 1)j$
- $\sum_{i=1}^{10} i(i - 1)$ and $\sum_{k=1}^{10} (k^2 - k)$

In the following exercises, use the rules for sums of powers of integers to compute the sums.

2. $\sum_{i=5}^{10} i$

3. $\sum_{i=5}^{10} i^2$

Suppose that $\sum_{i=1}^{100} a_i = 15$ and $\sum_{i=1}^{100} b_i = -12$. In the following exercises, compute the sums.

4. $\sum_{i=1}^{100} (a_i + b_i)$

5. $\sum_{i=1}^{100} (a_i - b_i)$

6. $\sum_{i=1}^{100} (3a_i - 4b_i)$

7. $\sum_{i=1}^{100} (5a_i + 4b_i)$

In the following exercises, use summation properties and formulas to rewrite and evaluate the sums.

8. $\sum_{k=1}^{20} 100(k^2 - 5k + 1)$

9. $\sum_{j=1}^{50} (j^2 - 2j)$

10. $\sum_{j=11}^{20} (j^2 - 10j)$

11. $\sum_{k=1}^{25} [(2k)^2 - 100k]$

Let L_n denote the left-endpoint sum using n subintervals and let R_n denote the corresponding right-endpoint sum. In the following exercises, compute the indicated left and right sums for the given functions on the indicated interval.

12. L_4 for $f(x) = \frac{1}{x-1}$ on $[2, 3]$

13. R_4 for $g(x) = \cos(\pi x)$ on $[0, 1]$

14. L_6 for $f(x) = \frac{1}{x(x-1)}$ on $[2, 5]$

15. R_6 for $f(x) = \frac{1}{x(x-1)}$ on $[2, 5]$

16. R_4 for $\frac{1}{x^2+1}$ on $[-2, 2]$

17. L_4 for $\frac{1}{x^2+1}$ on $[-2, 2]$

18. R_8 for $x^2 - 2x + 1$ on $[0, 2]$

19. L_8 for $x^2 - 2x + 1$ on $[0, 2]$

20. Compute the left and right Riemann sums— L_4 and R_4 , respectively—for $f(x) = (2 - |x|)$ on $[-2, 2]$. Compute their average value and compare it with the area under the graph of f .

21. Compute the left and right Riemann sums— L_6 and R_6 , respectively—for $f(x) = (3 - |3 - x|)$ on $[0, 6]$. Compute their average value and compare it with the area under the graph of f .

22. Compute the left and right Riemann sums— L_4 and R_4 , respectively—for $f(x) = \sqrt{4 - x^2}$ on $[-2, 2]$ and compare their values.

23. Compute the left and right Riemann sums— L_6 and R_6 , respectively—for $f(x) = \sqrt{9 - (x - 3)^2}$ on $[0, 6]$ and compare their values.

Express the following endpoint sums in sigma notation but do not evaluate them.

24. L_{30} for $f(x) = x^2$ on $[1, 2]$

25. L_{10} for $f(x) = \sqrt{4 - x^2}$ on $[-2, 2]$

26. R_{20} for $f(x) = \sin x$ on $[0, \pi]$

27. R_{100} for $\ln x$ on $[1, e]$

In the following exercises, graph the function then use a calculator or a computer program to evaluate the following left and right endpoint sums. If the two agree, say "neither."

28. [T] L_{100} and R_{100} for $y = x^2 - x + 3$ on the interval $[-1, 1]$
29. [T] L_{100} and R_{100} for $y = x^2$ on the interval $[0, 1]$
30. [T] L_{50} and R_{50} for $y = \frac{x+1}{x^2-1}$ on the interval $[2, 4]$
31. [T] L_{100} and R_{100} for $y = x^3 + 2$ on the interval $[-1, 1]$
32. [T] L_{50} and R_{50} for $y = \tan(x)$ on the interval $[0, \frac{\pi}{4}]$
33. [T] L_{100} and R_{100} for $y = e^{2x}$ on the interval $[-1, 1]$
34. Let t_j denote the time that it took Tejay van Garteren to ride the j th stage of the Tour de France in 2014. If there were a total of 21 stages, interpret $\sum_{j=1}^{21} t_j$.
35. Let r_j denote the total rainfall in Portland on the j th day of the year in 2009. Interpret $\sum_{j=1}^{31} r_j$.
36. Let d_j denote the hours of daylight and δ_j denote the increase in the hours of daylight from day $j - 1$ to day j in Fargo, North Dakota, on the j th day of the year. Interpret $d_1 + \sum_{j=2}^{365} \delta_j$.
37. To help get in shape, Joe gets a new pair of running shoes. If Joe runs 1 mi each day in week 1 and adds $\frac{1}{10}$ mi to his daily routine each week, what is the total mileage on Joe's shoes after 25 weeks?
38. The following table gives approximate values of the average annual atmospheric rate of increase in carbon dioxide (CO_2) each decade since 1960, in parts per million (ppm). Estimate the total increase in atmospheric CO_2 between 1964 and 2013.
39. The following table gives the approximate increase in sea level in inches over 20 years starting in the given year. Estimate the net change in mean sea level from 1870 to 2010.

Decade	Ppm/y
1964–1973	1.07
1974–1983	1.34
1984–1993	1.40
1994–2003	1.87
2004–2013	2.07

Table 5.2 Average Annual Atmospheric CO_2 Increase, 1964–2013 Source: <http://www.esrl.noaa.gov/gmd/ccgg/trends/>.

Starting Year	20-Year Change
1870	0.3
1890	1.5
1910	0.2
1930	2.8
1950	0.7
1970	1.1
1990	1.5

Table 5.3 Approximate 20-Year Sea Level Increases, 1870–1990 Source: <http://link.springer.com/article/10.1007%2Fs10712-011-9119-1>

40. The following table gives the approximate increase in dollars in the average price of a gallon of gas per decade since 1950. If the average price of a gallon of gas in 2010 was \$2.60, what was the average price of a gallon of gas in 1950?
41. The following table gives the percent growth of the U.S. population beginning in July of the year indicated. If the U.S. population was 281,421,906 in July 2000, estimate the U.S. population in July 2010.

Starting Year	10-Year Change
1950	0.03
1960	0.05
1970	0.86
1980	-0.03
1990	0.29
2000	1.12

Table 5.4 Approximate 10-Year Gas Price Increases, 1950–2000

Source: http://epb.lbl.gov/homepages/Rick_Diamond/docs/lbnl55011-trends.pdf.

Year	% Change/Year
2000	1.12
2001	0.99
2002	0.93
2003	0.86
2004	0.93
2005	0.93
2006	0.97
2007	0.96
2008	0.95
2009	0.88

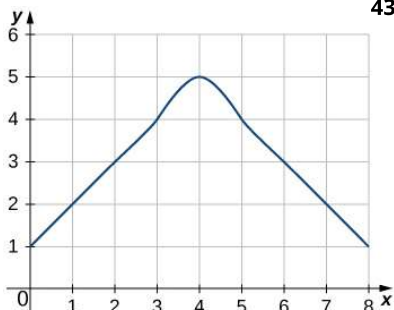
Table 5.5 Annual Percentage Growth of U.S. Population, 2000–2009 Source:

<http://www.census.gov/popest/data>.

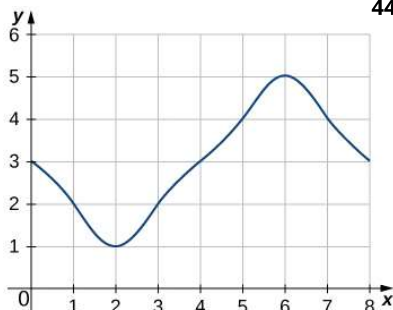
(Hint: To obtain the population in July 2001, multiply the population in July 2000 by 1.0112 to get 284,573,831.)

In the following exercises, estimate the areas under the curves by computing the left Riemann sums, L_8 .

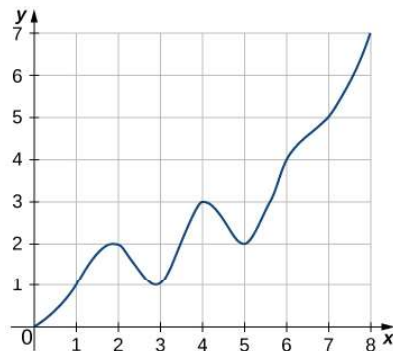
42.



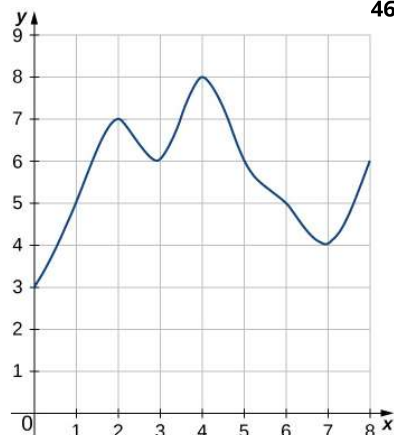
43.



44.



45.



46. [T] Use a computer algebra system to compute the Riemann sum, L_N , for $N = 10, 30, 50$ for $f(x) = \sqrt{1-x^2}$ on $[-1, 1]$.

47. [T] Use a computer algebra system to compute the Riemann sum, L_N , for $N = 10, 30, 50$ for $f(x) = \frac{1}{\sqrt{1+x^2}}$ on $[-1, 1]$.

48. [T] Use a computer algebra system to compute the Riemann sum, L_N , for $N = 10, 30, 50$ for $f(x) = \sin^2 x$ on $[0, 2\pi]$. Compare these estimates with π .

In the following exercises, use a calculator or a computer program to evaluate the endpoint sums R_N and L_N for $N = 1, 10, 100$. How do these estimates compare with the exact answers, which you can find via geometry?

49. [T] $y = \cos(\pi x)$ on the interval $[0, 1]$

50. [T] $y = 3x + 2$ on the interval $[3, 5]$

In the following exercises, use a calculator or a computer program to evaluate the endpoint sums R_N and L_N for $N = 1, 10, 100$.

51. [T] $y = x^4 - 5x^2 + 4$ on the interval $[-2, 2]$, which has an exact area of $\frac{32}{15}$

52. [T] $y = \ln x$ on the interval $[1, 2]$, which has an exact area of $2\ln(2) - 1$

53. Explain why, if $f(a) \geq 0$ and f is increasing on $[a, b]$, that the left endpoint estimate is a lower bound for the area below the graph of f on $[a, b]$.

54. Explain why, if $f(b) \geq 0$ and f is decreasing on $[a, b]$, that the left endpoint estimate is an upper bound for the area below the graph of f on $[a, b]$.

55. Show that, in general,

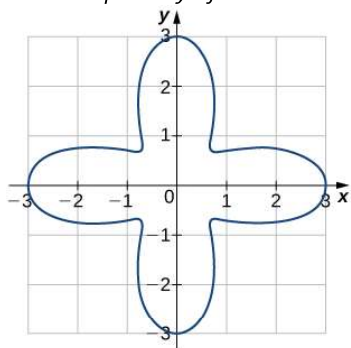
$$R_N - L_N = (b - a) \times \frac{f(b) - f(a)}{N}.$$

56. Explain why, if f is increasing on $[a, b]$, the error between either L_N or R_N and the area A below the graph of f is at most $(b - a) \frac{f(b) - f(a)}{N}$.

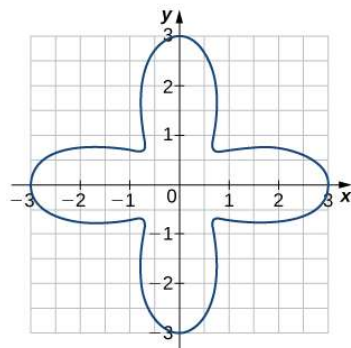
57. For each of the three graphs:

- Obtain a lower bound $L(A)$ for the area enclosed by the curve by adding the areas of the squares *enclosed completely* by the curve.
- Obtain an upper bound $U(A)$ for the area by adding to $L(A)$ the areas $B(A)$ of the squares *enclosed partially* by the curve.

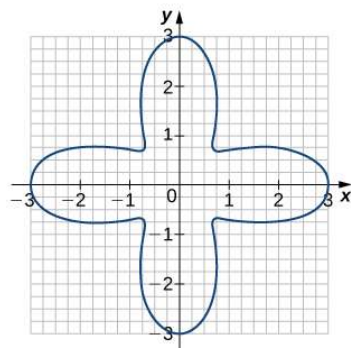
58. In the previous exercise, explain why $L(A)$ gets no smaller while $U(A)$ gets no larger as the squares are subdivided into four boxes of equal area.



Graph 1

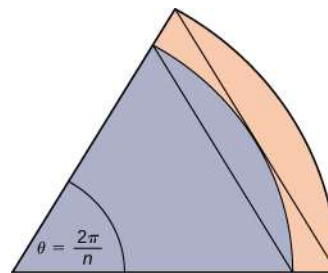


Graph 2



Graph 3

59. A unit circle is made up of n wedges equivalent to the inner wedge in the figure. The base of the inner triangle is 1 unit and its height is $\sin\left(\frac{2\pi}{n}\right)$. The base of the outer triangle is $B = \cos\left(\frac{\pi}{n}\right) + \sin\left(\frac{\pi}{n}\right) \tan\left(\frac{\pi}{n}\right)$ and the height is $H = B \sin\left(\frac{2\pi}{n}\right)$. Use this information to argue that the area of a unit circle is equal to π .



5.2 The Definite Integral

Learning Objectives

- 5.2.1 State the definition of the definite integral.
- 5.2.2 Explain the terms integrand, limits of integration, and variable of integration.
- 5.2.3 Explain when a function is integrable.
- 5.2.4 Describe the relationship between the definite integral and net area.
- 5.2.5 Use geometry and the properties of definite integrals to evaluate them.
- 5.2.6 Calculate the average value of a function.

In the preceding section we defined the area under a curve in terms of Riemann sums:

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x.$$

However, this definition came with restrictions. We required $f(x)$ to be continuous and nonnegative. Unfortunately, real-world problems don't always meet these restrictions. In this section, we look at how to apply the concept of the area under the curve to a broader set of functions through the use of the definite integral.

Definition and Notation

The definite integral generalizes the concept of the area under a curve. We lift the requirements that $f(x)$ be continuous and nonnegative, and define the definite integral as follows.

Definition

If $f(x)$ is a function defined on an interval $[a, b]$, the **definite integral** of f from a to b is given by

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x, \quad (5.8)$$

provided the limit exists. If this limit exists, the function $f(x)$ is said to be integrable on $[a, b]$, or is an **integrable function**.

The integral symbol in the previous definition should look familiar. We have seen similar notation in the chapter on [Applications of Derivatives](#), where we used the indefinite integral symbol (without the a and b above and below) to represent an antiderivative. Although the notation for indefinite integrals may look similar to the notation for a definite integral, they are not the same. A definite integral is a number. An indefinite integral is a family of functions. Later in this chapter we examine how these concepts are related. However, close attention should always be paid to notation so we know whether we're working with a definite integral or an indefinite integral.

Integral notation goes back to the late seventeenth century and is one of the contributions of Gottfried Wilhelm Leibniz, who is often considered to be the codiscoverer of calculus, along with Isaac Newton. The integration symbol $\int$ is an elongated S, suggesting sigma or summation. On a definite integral, above and below the summation symbol are the boundaries of the interval, $[a, b]$. The numbers a and b are x -values and are called the **limits of integration**; specifically, a is the **lower limit** and b is the **upper limit**. To clarify, we are using the word *limit* in two different ways in the context of the definite integral. First, we talk about the limit of a sum as $n \rightarrow \infty$. Second, the boundaries of the region are called the *limits of integration*.

We call the function $f(x)$ the **integrand**, and the dx indicates that $f(x)$ is a function with respect to x , called the **variable of integration**. Note that, like the index in a sum, the variable of integration is a dummy variable, and has no impact on the computation of the integral. We could use any variable we like as the variable of integration:

$$\int_a^b f(x) dx = \int_a^b f(t) dt = \int_a^b f(u) du$$

Previously, we discussed the fact that if $f(x)$ is continuous on $[a, b]$, then the limit $\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$ exists and is

unique. This leads to the following theorem, which we state without proof.

Theorem 5.1**Continuous Functions Are Integrable**

If $f(x)$ is continuous on $[a, b]$, then f is integrable on $[a, b]$.

Functions that are not continuous on $[a, b]$ may still be integrable, depending on the nature of the discontinuities. For example, functions continuous on a closed interval, apart from a finite number of jump discontinuities, are integrable.

It is also worth noting here that we have retained the use of a regular partition in the Riemann sums. This restriction is not strictly necessary. Any partition can be used to form a Riemann sum. However, if a nonregular partition is used to define the definite integral, it is not sufficient to take the limit as the number of subintervals goes to infinity. Instead, we must take the limit as the width of the largest subinterval goes to zero. This introduces a little more complex notation in our limits and makes the calculations more difficult without really gaining much additional insight, so we stick with regular partitions for the Riemann sums.

EXAMPLE 5.7**Evaluating an Integral Using the Definition**

Use the definition of the definite integral to evaluate $\int_0^2 x^2 dx$. Use a right-endpoint approximation to generate the Riemann sum.

✓ Solution

We first want to set up a Riemann sum. Based on the limits of integration, we have $a = 0$ and $b = 2$. For $i = 0, 1, 2, \dots, n$, let $P = \{x_i\}$ be a regular partition of $[0, 2]$. Then

$$\Delta x = \frac{b - a}{n} = \frac{2}{n}.$$

Since we are using a right-endpoint approximation to generate Riemann sums, for each i , we need to calculate the function value at the right endpoint of the interval $[x_{i-1}, x_i]$. The right endpoint of the interval is x_i , and since P is a regular partition,

$$x_i = x_0 + i\Delta x = 0 + i \left[\frac{2}{n} \right] = \frac{2i}{n}.$$

Thus, the function value at the right endpoint of the interval is

$$f(x_i) = x_i^2 = \left(\frac{2i}{n} \right)^2 = \frac{4i^2}{n^2}.$$

Then the Riemann sum takes the form

$$\sum_{i=1}^n f(x_i) \Delta x = \sum_{i=1}^n \left(\frac{4i^2}{n^2} \right) \frac{2}{n} = \sum_{i=1}^n \frac{8i^2}{n^3} = \frac{8}{n^3} \sum_{i=1}^n i^2.$$

Using the summation formula for $\sum_{i=1}^n i^2$, we have

$$\begin{aligned}
 \sum_{i=1}^n f(x_i) \Delta x &= \frac{8}{n^3} \sum_{i=1}^n i^2 \\
 &= \frac{8}{n^3} \left[\frac{n(n+1)(2n+1)}{6} \right] \\
 &= \frac{8}{n^3} \left[\frac{2n^3 + 3n^2 + n}{6} \right] \\
 &= \frac{16n^3 + 24n^2 + 8n}{6n^3} \\
 &= \frac{8}{3} + \frac{4}{n} + \frac{8}{6n^2}.
 \end{aligned}$$

Now, to calculate the definite integral, we need to take the limit as $n \rightarrow \infty$. We get

$$\begin{aligned}
 \int_0^2 x^2 dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x \\
 &= \lim_{n \rightarrow \infty} \left(\frac{8}{3} + \frac{4}{n} + \frac{8}{6n^2} \right) \\
 &= \lim_{n \rightarrow \infty} \left(\frac{8}{3} \right) + \lim_{n \rightarrow \infty} \left(\frac{4}{n} \right) + \lim_{n \rightarrow \infty} \left(\frac{8}{6n^2} \right) \\
 &= \frac{8}{3} + 0 + 0 = \frac{8}{3}.
 \end{aligned}$$

-
- ☒ 5.7 Use the definition of the definite integral to evaluate $\int_0^3 (2x - 1) dx$. Use a right-endpoint approximation to generate the Riemann sum.

Evaluating Definite Integrals

Evaluating definite integrals this way can be quite tedious because of the complexity of the calculations. Later in this chapter we develop techniques for evaluating definite integrals *without* taking limits of Riemann sums. However, for now, we can rely on the fact that definite integrals represent the area under the curve, and we can evaluate definite integrals by using geometric formulas to calculate that area. We do this to confirm that definite integrals do, indeed, represent areas, so we can then discuss what to do in the case of a curve of a function dropping below the x -axis.

EXAMPLE 5.8

Using Geometric Formulas to Calculate Definite Integrals

Use the formula for the area of a circle to evaluate $\int_3^6 \sqrt{9 - (x - 3)^2} dx$.

Solution

The function describes a semicircle with radius 3. To find

$$\int_3^6 \sqrt{9 - (x - 3)^2} dx,$$

we want to find the area under the curve over the interval $[3, 6]$. The formula for the area of a circle is $A = \pi r^2$. The area of a semicircle is just one-half the area of a circle, or $A = \left(\frac{1}{2}\right) \pi r^2$. The shaded area in [Figure 5.16](#) covers one-half of the semicircle, or $A = \left(\frac{1}{4}\right) \pi r^2$. Thus,

$$\begin{aligned}
 \int_3^6 \sqrt{9 - (x - 3)^2} \, dx &= \frac{1}{4} \pi (3)^2 \\
 &= \frac{9}{4} \pi \\
 &\approx 7.069.
 \end{aligned}$$

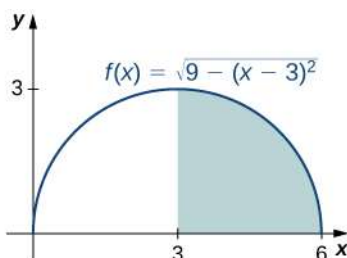


Figure 5.16 The value of the integral of the function $f(x)$ over the interval $[3, 6]$ is the area of the shaded region.

- ✓ 5.8 Use the formula for the area of a trapezoid to evaluate $\int_2^4 (2x + 3) \, dx$.

Area and the Definite Integral

When we defined the definite integral, we lifted the requirement that $f(x)$ be nonnegative. But how do we interpret “the area under the curve” when $f(x)$ is negative?

Net Signed Area

Let us return to the Riemann sum. Consider, for example, the function $f(x) = 2 - 2x^2$ (shown in Figure 5.17) on the interval $[0, 2]$. Use $n = 8$ and choose $\{x_i^*\}$ as the left endpoint of each interval. Construct a rectangle on each subinterval of height $f(x_i^*)$ and width Δx . When $f(x_i^*)$ is positive, the product $f(x_i^*) \Delta x$ represents the area of the rectangle, as before. When $f(x_i^*)$ is negative, however, the product $f(x_i^*) \Delta x$ represents the *negative* of the area of the rectangle. The Riemann sum then becomes

$$\sum_{i=1}^8 f(x_i^*) \Delta x = (\text{Area of rectangles above the } x\text{-axis}) - (\text{Area of rectangles below the } x\text{-axis})$$

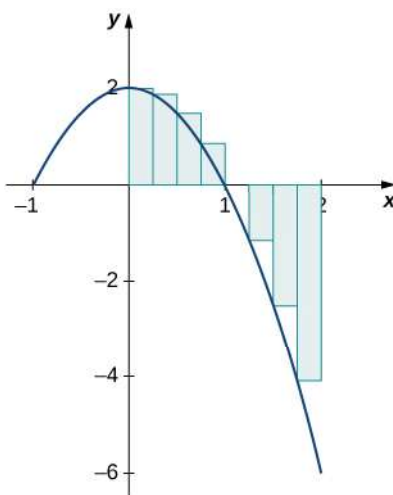


Figure 5.17 For a function that is partly negative, the Riemann sum is the area of the rectangles above the x -axis less the area of the rectangles below the x -axis.

Taking the limit as $n \rightarrow \infty$, the Riemann sum approaches the area between the curve above the x -axis and the x -axis, less the area between the curve below the x -axis and the x -axis, as shown in [Figure 5.18](#). Then,

$$\begin{aligned}\int_0^2 f(x) dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x \\ &= A_1 - A_2.\end{aligned}$$

The quantity $A_1 - A_2$ is called the **net signed area**.

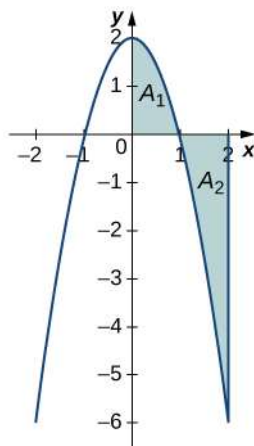


Figure 5.18 In the limit, the definite integral equals area A_1 less area A_2 , or the net signed area.

Notice that net signed area can be positive, negative, or zero. If the area above the x -axis is larger, the net signed area is positive. If the area below the x -axis is larger, the net signed area is negative. If the areas above and below the x -axis are equal, the net signed area is zero.

EXAMPLE 5.9

Finding the Net Signed Area

Find the net signed area between the curve of the function $f(x) = 2x$ and the x -axis over the interval $[-3, 3]$.

Solution

The function produces a straight line that forms two triangles: one from $x = -3$ to $x = 0$ and the other from $x = 0$ to $x = 3$ ([Figure 5.19](#)). Using the geometric formula for the area of a triangle, $A = \frac{1}{2}bh$, the area of triangle A_1 , above the axis, is

$$A_1 = \frac{1}{2}(3)(6) = 9,$$

where 3 is the base and $2(3) = 6$ is the height. The area of triangle A_2 , below the axis, is

$$A_2 = \frac{1}{2}(3)(6) = 9,$$

where 3 is the base and 6 is the height. Thus, the net area is

$$\int_{-3}^3 2x dx = A_1 - A_2 = 9 - 9 = 0.$$

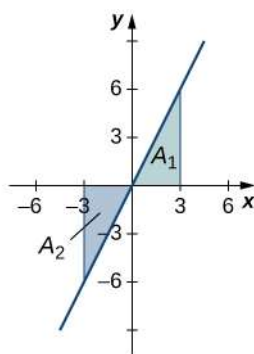
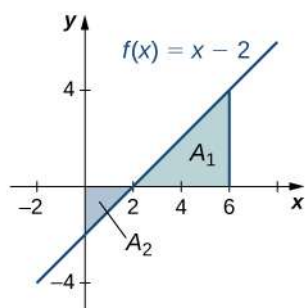


Figure 5.19 The area above the curve and below the x -axis equals the area below the curve and above the x -axis.

© **Analysis**

If A_1 is the area above the x -axis and A_2 is the area below the x -axis, then the net area is $A_1 - A_2$. Since the areas of the two triangles are equal, the net area is zero.

- ✓ 5.9 Find the net signed area of $f(x) = x - 2$ over the interval $[0, 6]$, illustrated in the following image.



Total Area

One application of the definite integral is finding displacement when given a velocity function. If $v(t)$ represents the velocity of an object as a function of time, then the area under the curve tells us how far the object is from its original position. This is a very important application of the definite integral, and we examine it in more detail later in the chapter. For now, we're just going to look at some basics to get a feel for how this works by studying constant velocities.

When velocity is a constant, the area under the curve is just velocity times time. This idea is already very familiar. If a car travels away from its starting position in a straight line at a speed of 70 mph for 2 hours, then it is 140 mi away from its original position (Figure 5.20). Using integral notation, we have

$$\int_0^2 70 dt = 140.$$

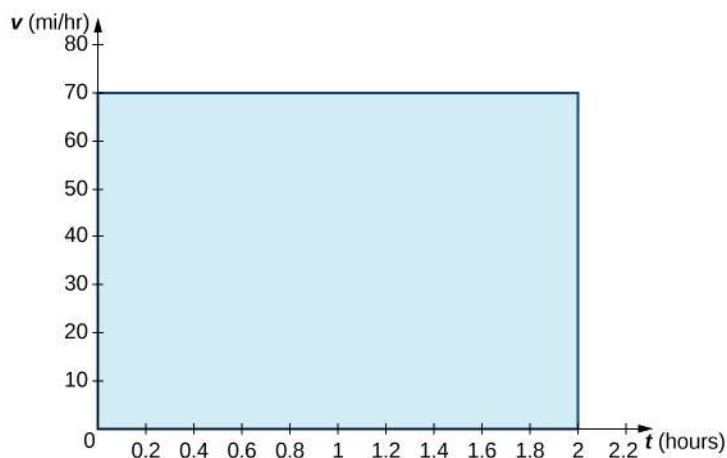


Figure 5.20 The area under the curve $v(t) = 70$ tells us how far the car is from its starting point at a given time.

In the context of displacement, net signed area allows us to take direction into account. If a car travels straight north at a speed of 60 mph for 2 hours, it is 120 mi north of its starting position. If the car then turns around and travels south at a speed of 40 mph for 3 hours, it will be back at its starting position ([Figure 5.21](#)). Again, using integral notation, we have

$$\begin{aligned} \int_0^2 60 dt + \int_2^5 -40 dt &= 120 - 120 \\ &= 0. \end{aligned}$$

In this case the displacement is zero.

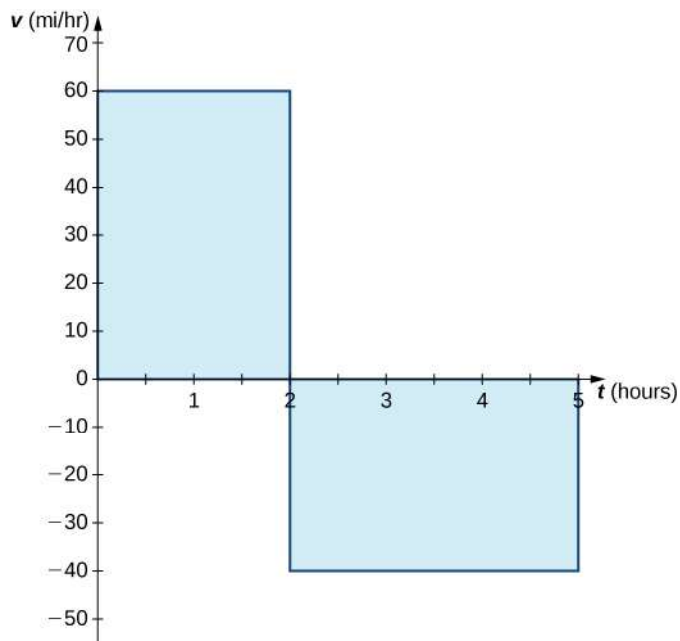


Figure 5.21 The area above the axis and the area below the axis are equal, so the net signed area is zero.

Suppose we want to know how far the car travels overall, regardless of direction. In this case, we want to know the area between the curve and the x-axis, regardless of whether that area is above or below the axis. This is called the **total area**.

Graphically, it is easiest to think of calculating total area by adding the areas above the axis and the areas below the axis (rather than subtracting the areas below the axis, as we did with net signed area). To accomplish this mathematically, we use the absolute value function. Thus, the total distance traveled by the car is

$$\begin{aligned}
 \int_0^2 |60| dt + \int_2^5 |-40| dt &= \int_0^2 60 dt + \int_2^5 40 dt \\
 &= 120 + 120 \\
 &= 240.
 \end{aligned}$$

Bringing these ideas together formally, we state the following definitions.

Definition

Let $f(x)$ be an integrable function defined on an interval $[a, b]$. Let A_1 represent the area between $f(x)$ and the x -axis that lies *above* the axis and let A_2 represent the area between $f(x)$ and the x -axis that lies *below* the axis. Then, the **net signed area** between $f(x)$ and the x -axis is given by

$$\int_a^b f(x) dx = A_1 - A_2.$$

The **total area** between $f(x)$ and the x -axis is given by

$$\int_a^b |f(x)| dx = A_1 + A_2.$$

EXAMPLE 5.10

Finding the Total Area

Find the total area between $f(x) = x - 2$ and the x -axis over the interval $[0, 6]$.

✓ Solution

Calculate the x -intercept as $(2, 0)$ (set $y = 0$, solve for x). To find the total area, take the area below the x -axis over the subinterval $[0, 2]$ and add it to the area above the x -axis on the subinterval $[2, 6]$ (Figure 5.22).

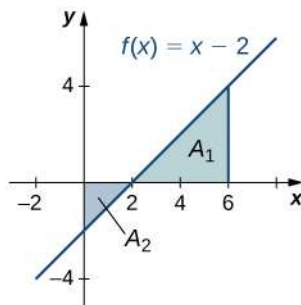


Figure 5.22 The total area between the line and the x -axis over $[0, 6]$ is A_2 plus A_1 .

We have

$$\int_0^6 |(x - 2)| dx = A_2 + A_1.$$

Then, using the formula for the area of a triangle, we obtain

$$A_2 = \frac{1}{2}bh = \frac{1}{2} \cdot 2 \cdot 2 = 2$$

$$A_1 = \frac{1}{2}bh = \frac{1}{2} \cdot 4 \cdot 4 = 8.$$

The total area, then, is

$$A_1 + A_2 = 8 + 2 = 10.$$

- ✓ 5.10 Find the total area between the function $f(x) = 2x$ and the x -axis over the interval $[-3, 3]$.

Properties of the Definite Integral

The properties of indefinite integrals apply to definite integrals as well. Definite integrals also have properties that relate to the limits of integration. These properties, along with the rules of integration that we examine later in this chapter, help us manipulate expressions to evaluate definite integrals.

Rule: Properties of the Definite Integral

1.

$$\int_a^a f(x) dx = 0 \quad (5.9)$$

If the limits of integration are the same, the integral is just a line and contains no area.

2.

$$\int_b^a f(x) dx = - \int_a^b f(x) dx \quad (5.10)$$

If the limits are reversed, then place a negative sign in front of the integral.

3.

$$\int_a^b [f(x) + g(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx \quad (5.11)$$

The integral of a sum is the sum of the integrals.

4.

$$\int_a^b [f(x) - g(x)] dx = \int_a^b f(x) dx - \int_a^b g(x) dx \quad (5.12)$$

The integral of a difference is the difference of the integrals.

5.

$$\int_a^b c f(x) dx = c \int_a^b f(x) dx \quad (5.13)$$

for constant c . The integral of the product of a constant and a function is equal to the constant multiplied by the integral of the function.

6.

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx \quad (5.14)$$

Although this formula normally applies when c is between a and b , the formula holds for all values of a , b , and c , provided $f(x)$ is integrable on the largest interval.

EXAMPLE 5.11

Using the Properties of the Definite Integral

Use the properties of the definite integral to express the definite integral of $f(x) = -3x^3 + 2x + 2$ over the interval $[-2, 1]$ as the sum of three definite integrals.

Solution

Using integral notation, we have $\int_{-2}^1 (-3x^3 + 2x + 2) dx$. We apply properties 3. and 5. to get

$$\begin{aligned}\int_{-2}^1 (-3x^3 + 2x + 2) dx &= \int_{-2}^1 -3x^3 dx + \int_{-2}^1 2x dx + \int_{-2}^1 2 dx \\ &= -3 \int_{-2}^1 x^3 dx + 2 \int_{-2}^1 x dx + \int_{-2}^1 2 dx.\end{aligned}$$

- ✓ 5.11 Use the properties of the definite integral to express the definite integral of $f(x) = 6x^3 - 4x^2 + 2x - 3$ over the interval $[1, 3]$ as the sum of four definite integrals.

EXAMPLE 5.12

Using the Properties of the Definite Integral

If it is known that $\int_0^8 f(x) dx = 10$ and $\int_0^5 f(x) dx = 5$, find the value of $\int_5^8 f(x) dx$.

☐ Solution

By property 6.,

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

Thus,

$$\begin{aligned}\int_0^8 f(x) dx &= \int_0^5 f(x) dx + \int_5^8 f(x) dx \\ 10 &= 5 + \int_5^8 f(x) dx \\ 5 &= \int_5^8 f(x) dx.\end{aligned}$$

- ✓ 5.12 If it is known that $\int_1^5 f(x) dx = -3$ and $\int_2^5 f(x) dx = 4$, find the value of $\int_1^2 f(x) dx$.

Comparison Properties of Integrals

A picture can sometimes tell us more about a function than the results of computations. Comparing functions by their graphs as well as by their algebraic expressions can often give new insight into the process of integration. Intuitively, we might say that if a function $f(x)$ is above another function $g(x)$, then the area between $f(x)$ and the x -axis is greater than the area between $g(x)$ and the x -axis. This is true depending on the interval over which the comparison is made. The properties of definite integrals are valid whether $a < b$, $a = b$, or $a > b$. The following properties, however, concern only the case $a \leq b$, and are used when we want to compare the sizes of integrals.

Theorem 5.2

Comparison Theorem

- i. If $f(x) \geq 0$ for $a \leq x \leq b$, then

$$\int_a^b f(x) dx \geq 0.$$

- ii. If $f(x) \geq g(x)$ for $a \leq x \leq b$, then

$$\int_a^b f(x) dx \geq \int_a^b g(x) dx.$$

- iii. If m and M are constants such that $m \leq f(x) \leq M$ for $a \leq x \leq b$, then

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a).$$

EXAMPLE 5.13**Comparing Two Functions over a Given Interval**

Compare $f(x) = \sqrt{1+x^2}$ and $g(x) = \sqrt{1+x}$ over the interval $[0, 1]$.

✓ Solution

Graphing these functions is necessary to understand how they compare over the interval $[0, 1]$. Initially, when graphed on a graphing calculator, $f(x)$ appears to be above $g(x)$ everywhere. However, on the interval $[0, 1]$, the graphs appear to be on top of each other. We need to zoom in to see that, on the interval $[0, 1]$, $g(x)$ is above $f(x)$. The two functions intersect at $x = 0$ and $x = 1$ (Figure 5.23).

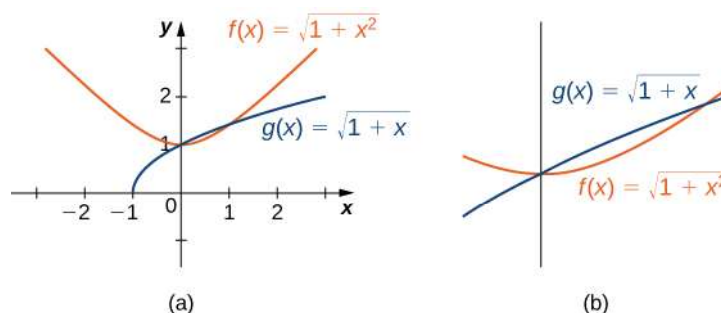


Figure 5.23 (a) The function $f(x)$ appears above the function $g(x)$ except over the interval $[0, 1]$ (b) Viewing the same graph with a greater zoom shows this more clearly.

We can see from the graph that over the interval $[0, 1]$, $g(x) \geq f(x)$. Comparing the integrals over the specified interval $[0, 1]$, we also see that $\int_0^1 g(x) dx \geq \int_0^1 f(x) dx$ (Figure 5.24). The thin, red-shaded area shows just how much difference there is between these two integrals over the interval $[0, 1]$.

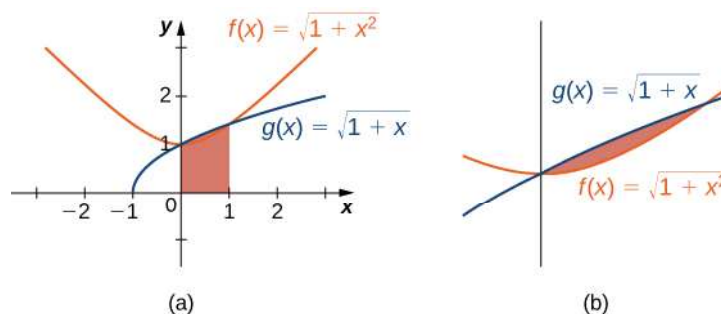


Figure 5.24 (a) The graph shows that over the interval $[0, 1]$, $g(x) \geq f(x)$, where equality holds only at the endpoints of the interval. (b) Viewing the same graph with a greater zoom shows this more clearly.

Average Value of a Function

We often need to find the average of a set of numbers, such as an average test grade. Suppose you received the following test scores in your algebra class: 89, 90, 56, 78, 100, and 69. Your semester grade is your average of test scores and you want to know what grade to expect. We can find the average by adding all the scores and dividing by the number of scores. In this case, there are six test scores. Thus,

$$\frac{89 + 90 + 56 + 78 + 100 + 69}{6} = \frac{482}{6} \approx 80.33.$$

Therefore, your average test grade is approximately 80.33, which translates to a B– at most schools.

Suppose, however, that we have a function $v(t)$ that gives us the speed of an object at any time t , and we want to find the object's average speed. The function $v(t)$ takes on an infinite number of values, so we can't use the process just described. Fortunately, we can use a definite integral to find the average value of a function such as this.

Let $f(x)$ be continuous over the interval $[a, b]$ and let $[a, b]$ be divided into n subintervals of width $\Delta x = (b - a)/n$. Choose a representative x_i^* in each subinterval and calculate $f(x_i^*)$ for $i = 1, 2, \dots, n$. In other words, consider each $f(x_i^*)$ as a sampling of the function over each subinterval. The average value of the function may then be approximated as

$$\frac{f(x_1^*) + f(x_2^*) + \cdots + f(x_n^*)}{n},$$

which is basically the same expression used to calculate the average of discrete values.

But we know $\Delta x = \frac{b-a}{n}$, so $n = \frac{b-a}{\Delta x}$, and we get

$$\frac{f(x_1^*) + f(x_2^*) + \cdots + f(x_n^*)}{n} = \frac{f(x_1^*) + f(x_2^*) + \cdots + f(x_n^*)}{\frac{(b-a)}{\Delta x}}.$$

Following through with the algebra, the numerator is a sum that is represented as $\sum_{i=1}^n f(x_i^*)$, and we are dividing by a fraction. To divide by a fraction, invert the denominator and multiply. Thus, an approximate value for the average value of the function is given by

$$\begin{aligned} \frac{\sum_{i=1}^n f(x_i^*)}{\frac{(b-a)}{\Delta x}} &= \left(\frac{\Delta x}{b-a} \right) \sum_{i=1}^n f(x_i^*) \\ &= \left(\frac{1}{b-a} \right) \sum_{i=1}^n f(x_i^*) \Delta x. \end{aligned}$$

This is a Riemann sum. Then, to get the *exact* average value, take the limit as n goes to infinity. Thus, the average value of a function is given by

$$\frac{1}{b-a} \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x = \frac{1}{b-a} \int_a^b f(x) dx.$$

Definition

Let $f(x)$ be continuous over the interval $[a, b]$. Then, the **average value of the function** $f(x)$ (or f_{ave}) on $[a, b]$ is given by

$$f_{\text{ave}} = \frac{1}{b-a} \int_a^b f(x) dx.$$

EXAMPLE 5.14

Finding the Average Value of a Linear Function

Find the average value of $f(x) = x + 1$ over the interval $[0, 5]$.

Solution

First, graph the function on the stated interval, as shown in [Figure 5.25](#).

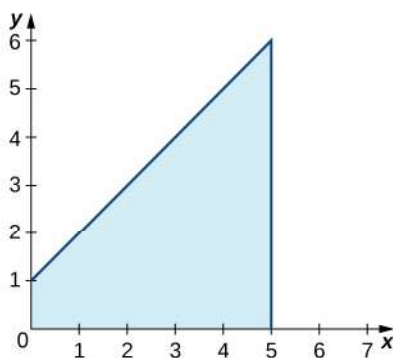


Figure 5.25 The graph shows the area under the function $f(x) = x + 1$ over $[0, 5]$.

The region is a trapezoid lying on its side, so we can use the area formula for a trapezoid $A = \frac{1}{2}h(a + b)$, where h represents height, and a and b represent the two parallel sides. Then,

$$\begin{aligned}\int_0^5 x + 1 dx &= \frac{1}{2}h(a + b) \\ &= \frac{1}{2} \cdot 5 \cdot (1 + 6) \\ &= \frac{35}{2}.\end{aligned}$$

Thus the average value of the function is

$$\frac{1}{5-0} \int_0^5 x + 1 dx = \frac{1}{5} \cdot \frac{35}{2} = \frac{7}{2}.$$

- ✓ 5.13 Find the average value of $f(x) = 6 - 2x$ over the interval $[0, 3]$.



SECTION 5.2 EXERCISES

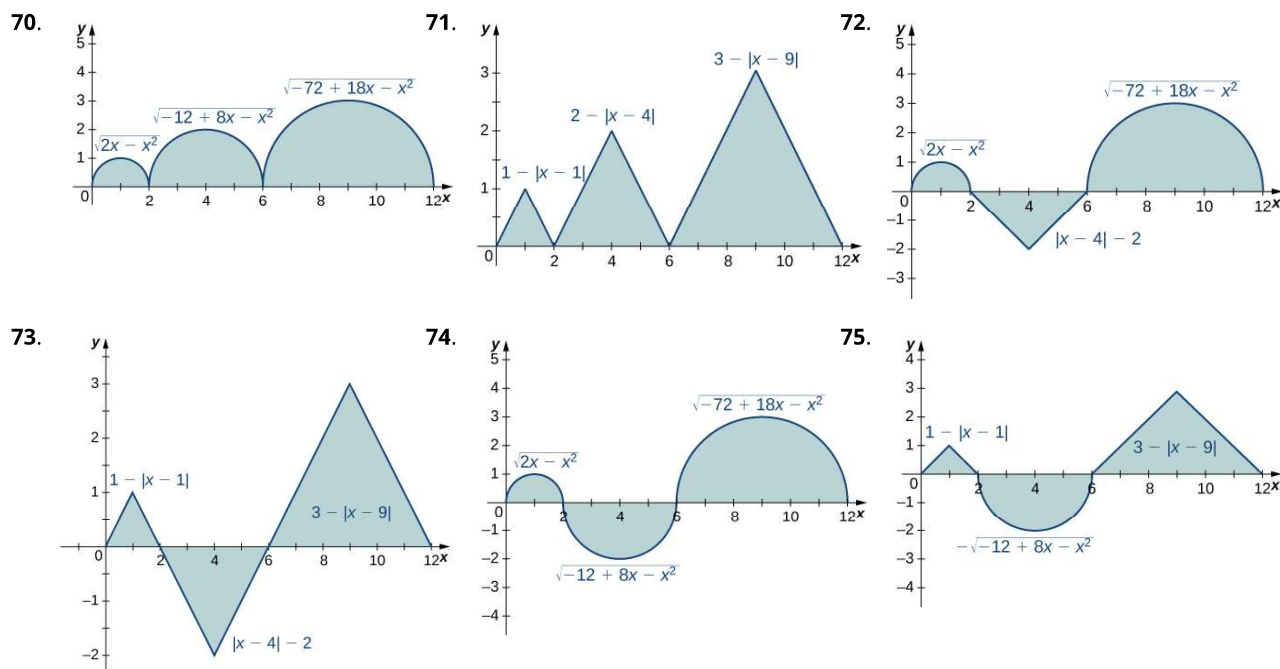
In the following exercises, express the limits as integrals.

60. $\lim_{n \rightarrow \infty} \sum_{i=1}^n (x_i^*) \Delta x$ over $[1, 3]$
61. $\lim_{n \rightarrow \infty} \sum_{i=1}^n (5(x_i^*)^2 - 3(x_i^*)^3) \Delta x$ over $[0, 2]$
62. $\lim_{n \rightarrow \infty} \sum_{i=1}^n \sin^2(2\pi x_i^*) \Delta x$ over $[0, 1]$
63. $\lim_{n \rightarrow \infty} \sum_{i=1}^n \cos^2(2\pi x_i^*) \Delta x$ over $[0, 1]$

In the following exercises, given L_n or R_n as indicated, express their limits as $n \rightarrow \infty$ as definite integrals, identifying the correct intervals.

64. $L_n = \frac{1}{n} \sum_{i=1}^n \frac{i-1}{n}$
65. $R_n = \frac{1}{n} \sum_{i=1}^n \frac{i}{n}$
66. $L_n = \frac{2}{n} \sum_{i=1}^n \left(1 + 2\frac{i-1}{n}\right)$
67. $R_n = \frac{3}{n} \sum_{i=1}^n \left(3 + 3\frac{i}{n}\right)$
68. $L_n = \frac{2\pi}{n} \sum_{i=1}^n 2\pi \frac{i-1}{n} \cos\left(2\pi \frac{i-1}{n}\right)$
69. $R_n = \frac{1}{n} \sum_{i=1}^n \left(1 + \frac{i}{n}\right) \log\left(\left(1 + \frac{i}{n}\right)^2\right)$

In the following exercises, evaluate the integrals of the functions graphed using the formulas for areas of triangles and circles, and subtracting the areas below the x -axis.



In the following exercises, evaluate the integral using area formulas.

76. $\int_0^3 (3-x) dx$ 77. $\int_2^3 (3-x) dx$ 78. $\int_{-3}^3 (3-|x|) dx$

79. $\int_0^6 (3-|x-3|) dx$ 80. $\int_{-2}^2 \sqrt{4-x^2} dx$ 81. $\int_1^5 \sqrt{4-(x-3)^2} dx$

82. $\int_0^{12} \sqrt{36-(x-6)^2} dx$ 83. $\int_{-2}^3 (3-|x|) dx$

In the following exercises, use averages of values at the left (L) and right (R) endpoints to compute the integrals of the piecewise linear functions with graphs that pass through the given list of points over the indicated intervals.

84. $\{(0, 0), (2, 1), (4, 3), (5, 0), (6, 0), (8, 3)\}$ over $[0, 8]$ 85. $\{(0, 2), (1, 0), (3, 5), (5, 5), (6, 2), (8, 0)\}$ over $[0, 8]$

86. $\{(-4, -4), (-2, 0), (0, -2), (3, 3), (4, 3)\}$ over $[-4, 4]$ 87. $\{(-4, 0), (-2, 2), (0, 0), (1, 2), (3, 2), (4, 0)\}$ over $[-4, 4]$

Suppose that $\int_0^4 f(x) dx = 5$ and $\int_0^2 f(x) dx = -3$, and $\int_0^4 g(x) dx = -1$ and $\int_0^2 g(x) dx = 2$. In the following exercises, compute the integrals.

88. $\int_0^4 (f(x) + g(x)) dx$ 89. $\int_2^4 (f(x) + g(x)) dx$ 90. $\int_0^2 (f(x) - g(x)) dx$

91. $\int_2^4 (f(x) - g(x)) dx$ 92. $\int_0^2 (3f(x) - 4g(x)) dx$ 93. $\int_2^4 (4f(x) - 3g(x)) dx$

In the following exercises, use the identity $\int_{-A}^A f(x) dx = \int_{-A}^0 f(x) dx + \int_0^A f(x) dx$ to compute the integrals.

94. $\int_{-\pi}^{\pi} \frac{\sin t}{1+t^2} dt$

(Hint: $\sin(-t) = -\sin(t)$)

95. $\int_{-\sqrt{\pi}}^{\sqrt{\pi}} \frac{t}{1+\cos t} dt$

In the following exercises, find the net signed area between $f(x)$ and the x -axis.

96. $\int_1^3 (2-x) dx$ (Hint: Look at the graph of f .)

97. $\int_2^4 (x-3)^3 dx$ (Hint: Look at the graph of f .)

In the following exercises, given that $\int_0^1 x dx = \frac{1}{2}$, $\int_0^1 x^2 dx = \frac{1}{3}$, and $\int_0^1 x^3 dx = \frac{1}{4}$, compute the integrals.

98. $\int_0^1 (1+x+x^2+x^3) dx$

99. $\int_0^1 (1-x+x^2-x^3) dx$

100. $\int_0^1 (1-x)^2 dx$

101. $\int_0^1 (1-2x)^3 dx$

102. $\int_0^1 \left(6x - \frac{4}{3}x^2\right) dx$

103. $\int_0^1 (7-5x^3) dx$

In the following exercises, use the [comparison theorem](#).

104. Show that

$$\int_0^3 (x^2 - 6x + 9) dx \geq 0.$$

105. Show that

$$\int_{-2}^3 (x-3)(x+2) dx \leq 0.$$

106. Show that

$$\int_0^1 \sqrt{1+x^3} dx \leq \int_0^1 \sqrt{1+x^2} dx.$$

107. Show that

$$\int_1^2 \sqrt{1+x} dx \leq \int_1^2 \sqrt{1+x^2} dx.$$

108. Show that

$$\int_0^{\pi/2} \sin t dt \geq \frac{\pi}{4}.$$

(Hint: $\sin t \geq \frac{2t}{\pi}$ over $[0, \frac{\pi}{2}]$)

109. Show that

$$\int_{-\pi/4}^{\pi/4} \cos t dt \geq \pi\sqrt{2}/4.$$

In the following exercises, find the average value f_{ave} of f between a and b , and find a point c , where $f(c) = f_{\text{ave}}$.

110. $f(x) = x^2, a = -1, b = 1$

111. $f(x) = x^5, a = -1, b = 1$

112. $f(x) = \sqrt{4-x^2}, a = 0, b = 2$

113. $f(x) = (3-|x|), a = -3, b = 3$

114. $f(x) = \sin x, a = 0, b = 2\pi$

115. $f(x) = \cos x, a = 0, b = 2\pi$

In the following exercises, approximate the average value using Riemann sums L_{100} and R_{100} . How does your answer compare with the exact given answer?

116. [T] $y = \ln(x)$ over the interval $[1, 4]$; the exact solution is $\frac{\ln(256)}{3} - 3$.

117. [T] $y = e^{x/2}$ over the interval $[0, 1]$; the exact solution is $2(\sqrt{e} - 1)$.

118. [T] $y = \tan x$ over the interval $[0, \frac{\pi}{4}]$; the exact solution is $\frac{2\ln(2)}{\pi}$.

119. [T] $y = \frac{x+1}{\sqrt{4-x^2}}$ over the interval $[-1, 1]$; the exact solution is $\frac{\pi}{6}$.

In the following exercises, compute the average value using the left Riemann sums L_N for $N = 1, 10, 100$. How does the accuracy compare with the given exact value?

120. [T] $y = x^2 - 4$ over the interval $[0, 2]$; the exact solution is $-\frac{8}{3}$.

121. [T] $y = xe^{x^2}$ over the interval $[0, 2]$; the exact solution is $\frac{1}{4}(e^4 - 1)$.

122. [T] $y = \left(\frac{1}{2}\right)^x$ over the interval $[0, 4]$; the exact solution is $\frac{15}{64\ln(2)}$.

- 123. [T]** $y = x \sin(x^2)$ over the interval $[-\pi, 0]$; the exact solution is $\frac{\cos(\pi^2) - 1}{2\pi}$.
- 124.** Suppose that $A = \int_0^{2\pi} \sin^2 t dt$ and $B = \int_0^{2\pi} \cos^2 t dt$. Show that $A + B = 2\pi$ and $A = B$.
- 125.** Suppose that $A = \int_{-\pi/4}^{\pi/4} \sec^2 t dt = \pi$ and $B = \int_{-\pi/4}^{\pi/4} \tan^2 t dt$. Show that $A - B = \frac{\pi}{2}$.
- 126.** Show that the average value of $\sin^2 t$ over $[0, 2\pi]$ is equal to $1/2$. Without further calculation, determine whether the average value of $\sin^2 t$ over $[0, \pi]$ is also equal to $1/2$.
- 127.** Show that the average value of $\cos^2 t$ over $[0, 2\pi]$ is equal to $1/2$. Without further calculation, determine whether the average value of $\cos^2(t)$ over $[0, \pi]$ is also equal to $1/2$.
- 128.** Explain why the graphs of a quadratic function (parabola) $p(x)$ and a linear function $\ell(x)$ can intersect in at most two points. Suppose that $p(a) = \ell(a)$ and $p(b) = \ell(b)$, and that $\int_a^b p(t) dt > \int_a^b \ell(t) dt$. Explain why $\int_c^d p(t) dt > \int_c^d \ell(t) dt$ whenever $a \leq c < d \leq b$.
- 129.** Suppose that parabola $p(x) = ax^2 + bx + c$ opens downward ($a < 0$) and has a vertex of $y = \frac{-b}{2a} > 0$. For which interval $[A, B]$ is $\int_A^B (ax^2 + bx + c) dx$ as large as possible?
- 130.** Suppose $[a, b]$ can be subdivided into subintervals $a = a_0 < a_1 < a_2 < \dots < a_N = b$ such that either $f \geq 0$ over $[a_{i-1}, a_i]$ or $f \leq 0$ over $[a_{i-1}, a_i]$. Set $A_i = \int_{a_{i-1}}^{a_i} f(t) dt$.
- Explain why $\int_a^b f(t) dt = A_1 + A_2 + \dots + A_N$.
 - Then, explain why $\left| \int_a^b f(t) dt \right| \leq \int_a^b |f(t)| dt$.
- 131.** Suppose f and g are continuous functions such that $\int_c^d f(t) dt \leq \int_c^d g(t) dt$ for every subinterval $[c, d]$ of $[a, b]$. Explain why $f(x) \leq g(x)$ for all values of x .
- 132.** Suppose the average value of f over $[a, b]$ is 1 and the average value of f over $[b, c]$ is 1 where $a < c < b$. Show that the average value of f over $[a, c]$ is also 1.
- 133.** Suppose that $[a, b]$ can be partitioned, taking $a = a_0 < a_1 < \dots < a_N = b$ such that the average value of f over each subinterval $[a_{i-1}, a_i] = 1$ is equal to 1 for each $i = 1, \dots, N$. Explain why the average value of f over $[a, b]$ is also equal to 1.
- 134.** Suppose that for each i such that $1 \leq i \leq N$ one has $\int_{i-1}^i f(t) dt = i$. Show that $\int_0^N f(t) dt = \frac{N(N+1)}{2}$.

135. Suppose that for each i such that $1 \leq i \leq N$ one has

$$\int_{i-1}^i f(t) dt = i^2. \text{ Show that } \int_0^N f(t) dt = \frac{N(N+1)(2N+1)}{6}.$$

138. If $\int_1^5 \sqrt{1+t^4} dt = 41.7133\dots$, what is $\int_1^5 \sqrt{1+u^4} du$?

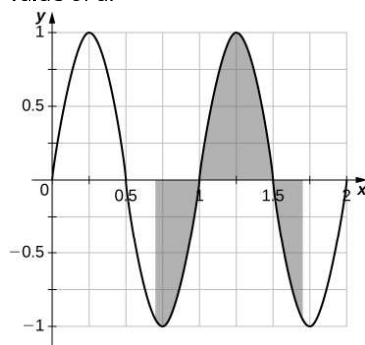
136. [T] Compute the left and right Riemann sums L_{10} and R_{10} and their average $\frac{L_{10}+R_{10}}{2}$ for $f(t) = t^2$ over $[0, 1]$. Given that $\int_0^1 t^2 dt = 0.\overline{33}$, to how many decimal places is $\frac{L_{10}+R_{10}}{2}$ accurate?

139. Estimate $\int_0^1 t dt$ using the left and right endpoint sums, each with a single rectangle. How does the average of these left and right endpoint sums compare with the actual value $\int_0^1 t dt$?

137. [T] Compute the left and right Riemann sums, L_{10} and R_{10} , and their average $\frac{L_{10}+R_{10}}{2}$ for $f(t) = (4-t^2)$ over $[1, 2]$. Given that $\int_1^2 (4-t^2) dt = 1.\overline{66}$, to how many decimal places is $\frac{L_{10}+R_{10}}{2}$ accurate?

140. Estimate $\int_0^1 t dt$ by comparison with the area of a single rectangle with height equal to the value of t at the midpoint $t = \frac{1}{2}$. How does this midpoint estimate compare with the actual value $\int_0^1 t dt$?

141. From the graph of $\sin(2\pi x)$ shown:
- Explain why $\int_0^1 \sin(2\pi t) dt = 0$.
 - Explain why, in general, $\int_a^{a+1} \sin(2\pi t) dt = 0$ for any value of a .



142. If f is 1-periodic ($f(t+1) = f(t)$), odd, and integrable over $[0, 1]$, is it always true that $\int_0^1 f(t) dt = 0$?
143. If f is 1-periodic and $\int_0^1 f(t) dt = A$, is it necessarily true that $\int_a^{1+a} f(t) dt = A$ for all A ?

5.3 The Fundamental Theorem of Calculus

Learning Objectives

- 5.3.1 Describe the meaning of the Mean Value Theorem for Integrals.
- 5.3.2 State the meaning of the Fundamental Theorem of Calculus, Part 1.
- 5.3.3 Use the Fundamental Theorem of Calculus, Part 1, to evaluate derivatives of integrals.
- 5.3.4 State the meaning of the Fundamental Theorem of Calculus, Part 2.
- 5.3.5 Use the Fundamental Theorem of Calculus, Part 2, to evaluate definite integrals.
- 5.3.6 Explain the relationship between differentiation and integration.

In the previous two sections, we looked at the definite integral and its relationship to the area under the curve of a function. Unfortunately, so far, the only tools we have available to calculate the value of a definite integral are geometric area formulas and limits of Riemann sums, and both approaches are extremely cumbersome. In this section we look at some more powerful and useful techniques for evaluating definite integrals.

These new techniques rely on the relationship between differentiation and integration. This relationship was discovered and explored by both Sir Isaac Newton and Gottfried Wilhelm Leibniz (among others) during the late 1600s and early